

Chapter 4

Boundary Ranks in Phonetics and Phonology

Abstract

The theory of syntax-prosody mapping presented in chapter 2 derives metrical presentations that define prosodic feet at various lines in a metrical grid. The number of the grid line corresponds to the strength of the perceived boundary separating feet at a given line. This chapter reviews how well this representation reflects the phonetics and phonology of prosodic structure, and argues that it correctly captures that (i) prosodic structure reflects the expected boundary ranks, and (ii) that the mapping between syntax and phonology is recursive in nature and not tied to specific syntactic categories.

4.1	The Representation of Prosodic Structure	132
4.2	Quantitative Evidence	133
4.3	Qualitative Evidence	143

4.1 The Representation of Prosodic Structure

The prosodic facts reported in chapter 2 are based on intuitions on prosodic grouping based on boundary strength. They are used to motivate a relational metrical grid, first proposed in Liberman (1975) and developed in many subsequent studies (e.g. Prince (1983), Hayes (1984), Selkirk (1984)). I am assuming an adapted version of the bracketed version of the metrical grid (cf. Halle and Vergnaud, 1987, Idsardi, 1992), which organizes the phonological representation of an expression recursively into feet.

(1) Relational Metrical Grid

	×		×		×		×		4
	×		×		×		×		3
	×		×		×		×		2
	×		×		×		×		1
	A		B		C		D		

The grid encodes grouping by virtue of foot boundaries of different strength: The strength of a foot boundary at line *i* in the grid is perceived as stronger than the strength of a foot boundary at line *i*-1. This chapter looks at the phonetics and phonology of prosody and how it relates to these boundary ranks.

There are various quantitative effects that suggest that a recursive hierarchy of boundary ranks makes the correct predictions. One set of facts relates to durational effects: domain-initial strengthening and domain-final lengthening are both cumulative effects that correlate with boundary rank. A second effect regards pitch-accent scaling. While whether or not a pitch accent is down-stepped may be a categorical effect, the precise scaling of the pitch value requires a recursive representation that reflects nested layers of grouping, as expected based on the recursive metrical grid representation. These quantitative effects will be discussed in section 4.2.

The grid in (1) represents hierarchically organized feet. Feet and their boundaries fall into different categories by virtue of being on different grid lines. However, the theory does not categorize these feet into prosodic constituents with different labels, as is usually assumed in the literature on the ‘prosodic hierarchy’. An example of a prosodic hierarchy is given in (2):

- (2) Prosodic hierarchy (Selkirk, 1986, 384):
- | | |
|--|-----|
| (-----) | Utt |
| (-----)(-----) | IPh |
| (-----)(-----)(-----) | PPh |
| (-----)(-----)(-----)(-----)(-----) | PWd |
| (--)(-----)(-----)(--)(-----)(-----)(--)(-----) | Ft |
| (--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--) | Syl |

While the theory is compatible with an implementation of the different boundary ranks that distinguishes prosodic domains of different types, the central departure from earlier work on the mapping between syntax and prosody is that the category of the prosodic domains cannot be directly related to syntactic categories such as $\bar{X}$ vs. XP . The issue and its implications are discussed in section 4.3.

The discussion of this chapter forms the basis for an experimental investigation of the relational view of metrical structure. The results will be discussed in chapter 5.

4.2 Quantitative Evidence

Prosodic structure has effects on the duration of segmental gestures. This section presents evidence that domain-initial strengthening and pre-boundary lengthening are cumulative effects, which correlate with the boundary ranks provided by the metrical structure proposed here.

4.2.1 Durational Effects and Boundary Strength Scaling

Durational effects were identified in Lehiste (1973) as the most reliable in disambiguating syntactic structures based on their bracketing. This section reviews evidence from pre-boundary lengthening, pauses, and domain-initial strengthening.

The coordinate structures in (3) were among the data set Lehiste (1973) investigated, and was one of the structures for which listeners scored best in identifying the correct bracketing in a listening task.

- (3) a. ((Steve or Sam) and Bob) will come.

- b. (Steve or (Sam and Bob)) will come.

Lehiste (1973, 117) found that the length of the underlined parts were longer if it contained a constituent boundary—that is if the higher-level bracket cuts into this constituent as in (3q). Sometimes, but not always, there was laryngealization and/or a pause at the constituent break. She also notes that ‘Sam and’ in the structure (3b) was ‘drawled’, i.e. reduced and less carefully pronounced.¹ These durational effects observed by Lehiste (1973) can be related directly to boundary ranks.

Boundary Ranks affect duration. The strength of a boundary is reflected by pre-boundary lengthening of the material preceding it. Klatt (1975) showed that segments are lengthened preceding boundaries, even in the absence of pauses. Pre-boundary lengthening affects mainly the material directly adjacent to a prosodic boundary, but also the main stress syllable according to Shattuck-Hufnagel (2005). It has been shown to correlate closely with the strength of the following boundary (Wightman et al., 1992, Price et al., 1991, Shattuck-Hufnagel and Turk, 1996).²

Byrd and Saltzman (1998), Cho and Keating (2001), and Byrd and Saltzman (2003) concluded that the degree of lengthening at particular boundaries depends on the strength of the boundaries, but that the difference is one of degree and not one of *type* of effect. In other words, durational effects give evidence for a relative notion of boundary strength but not of different types of boundaries.

The metrical grid assigned here provides a measure of the strength of boundaries: the boundary rank. The boundary ranks differentiate prosodic domains on the discrete scale defined by the metrical grid representation. The degree of domain-final lengthening in coordinate structure will be shown in chapter 5 to correlate with the

¹In a follow-up study (Lehiste et al., 1976), the sentences were manipulated in order to verify whether segmental duration (as opposed to pauses and pitch) was one of the cues used by listeners in assigning a syntactic structure. Pitch was set constant at 100 Hertz for the sentences, and duration was manipulated. The results showed that durational cues indeed determined the reading listeners assigned to the structures.

²Some researchers (e.g. Martin (1971)) interpret pre-boundary lengthening as a ‘slow down’, and speculate that that this reflects the extra-time needed by the speaker to compose the next phrase. I will return to this point in chapter 5.

boundary ranks predicted by the metrical grids proposed here.

The boundary ranks assigned here are relative. A boundary is stronger or weaker than other boundaries depending on its rank. The metrical grid does not specify an absolute strength or category of a boundary. The theory is then able to account for quantitative phonetic effects that distinguish different prosodic boundary ranks—it does not distinguish different *qualitative* categories of prosodic boundaries.

There is phonetic evidence that boundaries are indeed scaled relative to each other. Carlson et al. (2001) show evidence that the *relative* boundary strength is used to disambiguate attachment ambiguities.

In chapter 5, a model is proposed in which *all* boundaries are phonetically realized by scaling the strength of a boundary relative to already produced boundaries earlier in the utterance.

(4) BOUNDARY STRENGTH SCALING

The strength of a boundary is scaled relative to d boundaries produced earlier. It must be realized such that it is perceived with the appropriate assigned boundary rank compared to boundaries of different or equal rank realized before.

Durational cues such as pre-boundary lengthening are used to achieve the correct boundary strength scaling. The very first constituent in an utterance sets the benchmark for the realization of later constituents. This relative scaling of boundary ranks is compatible with an unlabeled grid, and shows that indeed the organization of prosodic feet into a hierarchy may be relative and not categorical. Experimental evidence for this model is presented in chapter 5.

Another durational correlate of boundary strength are *pauses*. Grosjean and Collins (1979) asked subjects to read test sentences at various speech rates. They added the overall length of pauses produced between each pair of adjacent words as a measure of the grouping of words. These pause durations were then used to derive binary tree structures in the following way: the two words separated by the shortest pause duration were grouped together under one node, and

then treated as one word afterwards; this procedure was continued until a complete structure was derived. Crucial in deriving this tree representation was the *relative* duration of pauses between words. The derived tree structure matched closely listener's intuitions about grouping of the sentences, which were independently elicited. Gee and Grosjean (1983) gave a prosodic interpretation to the derived tree structures, again based on relative degrees of pause duration, as opposed to categorical criteria that identify prosodic constituents of a particular kind.

Pre-Boundary lengthening and perceived pauses are closely related. Martin (1970, 1971) found that listeners report hearing pauses although there were no unfilled pauses in the signal. In those cases, there was evidence for syllable lengthening just before the perceived juncture. The syllable before the juncture was longer compared to the syllable following the juncture. Conversely, for those pauses that were present in the signal but not detected by listeners, syllables before the boundary were *not* lengthened—syllable lengthening was thus the dominant cue.

Finally, domain-initial strengthening is also a durational effect that reflects boundary strength. Fougeron and Keating (1997), Lavoie (2001), Cho (2002), Keating et al. (2003), Keating (to appear) discuss evidence that the phonetic realization of segments depends on their position in the prosodic structure. In particular, they argue that articulatory strengthening occurs at the beginning of prosodic domains, and that this strengthening quantitatively increases cumulatively with the strength of the prosodic boundary.

One type of experiment conducted in order to show this effect uses electropalatography to measure linguopalatal contact, which is a reflex of the degree of oral constriction. The examples investigated in Fougeron and Keating (1997) were arithmetic expressions, whose bracketing is of course very similar to the bracketing one finds in coordinate structures:

- (5) a. $89 + 89 + 89 + 89 = \text{a lot.}$
- b. $(89 + 89) * (89 + 89) = \text{a lot.}$
- c. $89 * (89 + 89 + 89) = \text{a lot.}$

d. $(89 + 89 + 89) * 89 = \text{a lot.}$

Subjects were asked to read statements in reiterant speech that were based on the bracketing modeled by the arithmetic formulae. The following statement is equivalent in bracketing to (5d):

(6) (nonono no nonono no nonono) no nonono equals a lot

The results of the experiment showed (i) that nasals are articulated more strongly at the beginning of domains compared to domain-medially, and (ii) that they are articulated more strongly at the beginning of higher-order prosodic constituents compared to lower order prosodic constituents.

Similar results were reported from an acoustic study of VOT in Korean in Jun (1993), based on which Jun suggests that there is hierarchy of strength which can be defined in relation to the position in the prosodic structure.

Fougeron and Keating (1997) propose that phonetic strengthening is cumulative and continuously increases with the strength of the boundary. It is not tied to particular prosodic domains (say the prosodic word or the intonational phrase). Cho and Keating (2001), Keating (to appear) explicitly argue that domain-initial strengthening is continuous, and does not lend support for a fixed category of a particular number of categorically distinct types of prosodic constituents.

Keating et al. (2003) illustrate evidence that domain-initial strengthening occurs cross-linguistically, independent of the particular prosodic system of the language. Thus languages that are not stress-based such as English, e.g. French, Korean, and Taiwanese, show very similar patterns of domain-initial strengthening.

Not only are segments strengthened at the beginning of prosodic domains, segments can also be inserted to strengthen the beginning of a domain. Dilley et al. (1996) show evidence that glottal stop insertion is more likely at stronger prosodic domain breaks compared to weaker boundaries.

Based on the evidence in the phonetics literature, we can conclude that domain-initial strengthening and pre-boundary lengthening are cumulative effects, which can be related to the boundary ranks the metrical grid structures derived here provide.

Boundaries of different strengths differ in the degree of the lengthening/strengthening they induce, but they are not differentiated in the type of effect they have.

The evidence from studies on domain final lengthening, pause realization, and domain-initial strengthening is compatible with a prosodic representation that only distinguishes relative boundary ranks, and not categorically distinguished prosodic constituents.

The experiment presented in chapter 5 provides more evidence that the phonetic implementation of prosodic structure is indeed relative, and that the strength of prosodic boundaries is scaled relative to earlier boundaries in the utterance.

4.2.2 Pitch Accent Scaling

The metrical structure does not only play a role in determining *where* accents are placed. It also influences *how* they are realized. Ladd (1988) and Fery and Truckenbrodt (2004) show evidence from pitch accent scaling in English and German respectively. The following type of coordination structure was investigated:

- (7) a. A but (B and C)
- b. (A and B) but C

The following are example sentences illustrating the data:

- (8) a. Ryan has a lot more money, (but Warren is a stronger campaigner and Allen has more popular policies.
- b. Allen is a stronger campaigner, and Ryan has more popular policies, but Warren has a lot more money.

The pitch accent scaling distinguishes the two types of structure. In structures of type (7a), conjunct C is downstepped relative to B, and B in turn is downstepped relative to A. The pitch level goes down from accent to accent. In structures of type (7b) on the other hand, C and B are at about the same level³, but both are set to a lower

³In fact, C is scaled a little higher in pitch than B, a fact discussed in detail in Fery and Truckenbrodt (2004).

pitch compared to A. This contrast in pitch scaling requires a prosodic representation that encodes the syntactic difference between (7a) and (7b).

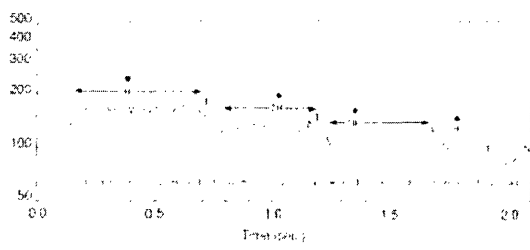
Ladd proposes to explain the difference in pitch scaling by a hierarchically metrical representation that allows a recursive nesting of intonational phrases. Within each level of coordinate structure, conjuncts are downstepped relative to the preceding conjunct. In structures of type (7b), conjunct C is downstepped relative to the first conjunct (A and B). This has the effect that the pitch of an accent in C is lower than the maximal pitch in (A and B), but it is not lower than the pitch in a preceding conjunct, in this case B.

In van den Berg et al. (1992), evidence is presented for scaling from coordinate structures involving place names, and show further evidence for Ladd's idea about how pitch scaling reflects hierarchical structure. The accent within each conjunct is downstepped relative to the preceding one ('en' is the Dutch for 'and'):

- (9) Harlem, Rommeldam, Harderwijk en Den Helder.

The pitch scaling is illustrated in the following figure:

- (10) Downstepping in 'Flat' Coordinate Structure



When a coordinate structure is further bracketed, then the pitch scaling reflects this bracketing:

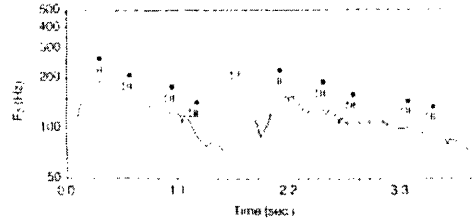
- (11) Downstepping in a Nested Structure

(Merel, Nora, Leo, Remy), en (Nelie, Mary, Leendert, Mona en Lorna).

The pitch accents on the conjuncts *within* each of the more embedded brackets is as in the unembedded structure in (9). Each accent is downstepped relative to a preceding one. But there is a further downstepping effect, that affects *the entire cycle* (Nelie,

Mary, Leendert, Mona en Lona): It is downstepped in pitch range relative to the first conjunct (Merel, Nora, Leo, Remy).

(12) Pitch Scaling in Nested Coordinate Structure



The generalization about pitch scaling has to be recursively stated: downstepping exists at different levels of embedding. This is captured in van den Berg et al. (1992), who propose that accents are downstepped relative to preceding accents, and in addition, entire domains containing accents can be downstepped relative to preceding domains. How can we state the generalization based on the prosodic representations used here?

Consider the representation assigned here for structures with the bracketing in (7):

(76) Metrical Grid

a. A or (B and C)

b. (A or B) and C

	×		×		×			×		×		×	
	×		×		×			×		×		×	
	A		B		C			A		B		C	

In deciding how to scale the accent on C, it matters how deeply embedded C is. I define a notion of embeddedness of accents by introducing the concept of a ‘minimal foot’.

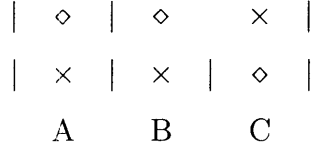
(13) Minimal Foot

The minimal foot of a grid mark is the highest foot in the column in which the grid mark is adjacent to a preceding foot boundary.

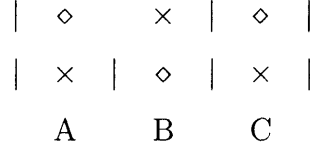
I mark the minimal foot for each top line grid mark by the symbol $\diamond$:

(14) Metrical Grid

a. A or (B and C)



b. (A or B) and C



I will assume with van den Berg et al. (1992) that each foot has a pitch register level. This is the pitch value to which the first H* that it contains is scaled, if there is one. The generalization about downstepping can now be stated as follows:

(15) Pitch Accent Scaling

The pitch level of a minimal foot is downstepped relative to the pitch level of the foot that precedes it on the same line.

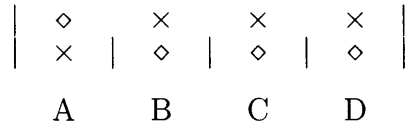
H* pitch accents are realized at the pitch level of the current minimal foot. This has the effect that if H* pitch accents are placed, they are downstepped relative to the preceding pitch level of the closest minimal foot (marked by ◇) at the same grid line or a higher line.

Consider the minimal foot corresponding to the grid mark of C in the two structures in (76). The convention on pitch accent scaling states that C is scaled relative to the accent on B in (76a), but is scaled relative to the pitch level of the foot containing (A and B) in (76b). Since the pitch level of this foot is the same as the level to which the accent on A is scaled, this effectively means that C is downstepped relative to A, just as proposed in Ladd (1988).⁴

⁴How can we motivate the notion ‘minimal foot’? The pitch register level of a prosodic domain is necessary in order to phonetically implement an accent. If the pitch scaling convention here is correct, then the information about how to realize the accents within the minimal foot phonetically is available once this foot is completed, since only the preceding minimal foot is necessary to scale its pitch level correctly. The amount of information that has to be computed (or anticipated) before a constituent can be pronounced is the minimal foot, so maybe minimal feet are planning units for speech production.

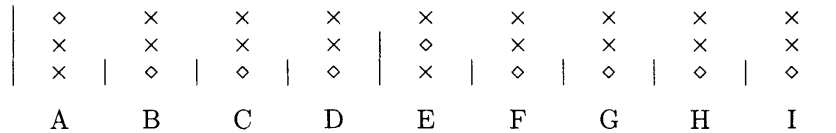
This statement about pitch scaling also works for the unembedded structure in (9):

(16) Prosodic Structure for (9)



The pitch level in each conjunct is downstepped relative to the immediately preceding foot. This has the effect that H* placed on those conjuncts will be down-stepped in pitch level from left to right. The metrical structure assigned to the nested structure in (11) looks as follows:

(17) Prosodic Structure for (11)



Pitch accent scaling has been used in Ladd (1988) to argue for a recursive nesting of prosodic categories with the same phonological ‘label’, the intonational phrase. The representation used here provides such a recursive notation, albeit one without any labels at all. More evidence for recursive nesting can be found in the recursive accent system of Tiberian Hebrew used in the Masoretic Bible (Dresher, 1994). Dresher argues that the accents annotate prosodic structure, and relate directly to regularities in segmental phonology and in the distribution of pausal forms.

The grid encodes the information necessary to state the generalization about pitch scaling observed in Ladd (1988) and van den Berg et al. (1992). More evidence for pitch scaling is discussed in Kubozono (1989), Kubozono (1992), and Fery and Truckenbrodt (2004). Further arguments in favor of a hierarchical representation of metrical structure and a model of how accents can be implemented based on it are discussed in Dilley (2005).

To summarize, there is quantitative evidence that suggests that boundaries of different ranks are distinguished by gradient phonetic correlates in duration and pitch

value. If all correlates of prosody domains were gradient, then a relational and recursive representation such as the metrical grid would be entirely sufficient to capture prosodic structure. The next section will review evidence for categorical distinctions between prosodic domains, and how they can be made compatible with the recursive approach pursued here.

4.3 Qualitative Evidence

The phonological literature on prosody often assumes a ‘prosodic hierarchy’ of phonological domains. This was proposed in (Selkirk, 1978, 1980, 1981) and was extended in Nespor and Vogel (1982, 1986). The metrical grid derived in chapter 2 is purely relational, just as the metrical grid proposed in (Liberman, 1975, Libermann and Prince, 1977, Prince, 1983, Selkirk, 1984). This section discusses evidence for categorical distinctions between prosodic domains, and how they can be made compatible with the proposal in chapter 2.

The main point will be that the theory is compatible with categorical distinctions between prosodic domains, but the theory is not categorical with a theory that tries to tie specific prosodic labels to specific syntactic categories (e.g. $\bar{X}$ vs. XP). Instead, the mapping between syntax and phonology is relational, even if the implementation of the relevant ranks in phonology may distinguish prosodic domains with different labels.

4.3.1 Designated Categories

The prosodic hierarchy has been key in many subsequent studies of prosody (cf. Selkirk, 1986, Hayes, 1989, Inkelas and Zec, 1990, Selkirk, 1995b). It consists of a layered prosodic structure in which each line bears a different label. Different authors have made different proposals with respect to the exact number and label of the categories. The following is one illustration, repeated from (2):

- (2) Prosodic hierarchy (Selkirk, 1986, 384):

(-----)	Utt
(-----)(-----)	IPh
(-----)(-----)(-----)	PPh
(-----)(-----)(-----)(-----)(-----)	PWd
(--)(-----)(-----)(--)(-----)(-----)(--)(-----)	Ft
(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)(--)	Syl

The different types of prosodic domains were originally identified by phonological rules that apply only within certain domains but not others. Work on the tonology of prosodic domains (Pierrehumbert, 1980, Beckman and Pierrehumbert, 1986) lead to a second set of criteria to identify types of prosodic domains, namely particular pitch accents and boundary tones. These criteria were developed into an annotation system to transcribe intonation, the ToBI-system (Silverman et al., 1992). The international criteria and the criteria based on phonological rule application are assumed to converge on the same categories, and several studies have argued that there is indeed such a correlation (i.a. Jun, 1998).

A central assumption in much of the literature on prosodic phonology is that a nesting of prosodic constituents bearing the same label is impossible (Selkirk, 1986). This restriction has been demoted to a violable rather than an inviolable constraint in the context of Optimality Theory in Selkirk (1995a, 443) and Truckenbrodt (1995) (C^i stands for a prosodic category with label i):

$$(18) \quad \textit{Nonrecursivity:} \text{ no } C^i \text{ dominates } C^j, j = i.$$

The metrical grid derived in chapter 2 on the other hand is purely relational, just as the metrical grid originally proposed in Liberman (1975) and Libermann and Prince (1977). Liberman (1975, 73) describes the idea behind a relational metrical grid as follows:

”A metrical grid is a pattern which subdivides intervals of time, in the way that is familiar from musical notation; [...] The Grid, like time, is in principle infinite – we can pick any place in it to start, and have available

sequences forward and backward, or up and down in the hierarchy, as we choose.”

The representation used here is a version of the bracketed grid (Halle and Vergnaud, 1987). Prosodic feet are ranked relative to each other by virtue of which line in the grid they are part of, but not by a difference in categorial label. The representation thus captures some but not all of the information encoded in the prosodic hierarchy.

(1) Relational Metrical Grid

	×		×		×		×	
	×		×		×		×	
	×		×		×		×	
	×		×		×		×	
	A		B		C		D	

What the metrical structure used here fails to assign are *labels* of the grid-lines. Prosodic feet are different in status only by virtue of being on different lines in the grid.

Since Liberman’s proposal, a number of studies have used an unlabeled recursive metrical structure to encode prosody, e.g. Selkirk (1984), Halle and Vergnaud (1987), Idsardi (1992), Cinque (1993), Arregi (2002) and Ishihara (2003). Selkirk (1984, 26ff) presents arguments in favor of reducing the prosodic hierarchy to the intonation phrase and the syllable, and represent prosody otherwise by a relational grid representation.

The representation used here is compatible with a mapping of the output of the algorithm to a fixed hierarchy of categorial prosodic categories. For example, it is possible that the syntax-to-phonology mapping principles give us the unlabeled grid, which is then mapped to a labeled grid in a second stage of phonological implementation. The prosodic hierarchy would then be the way the relative boundary distinctions provided by the syntax-phonology mapping are formally implemented by the phonological component.

What the theory presented here is *not* compatible with is that specific syntactic constituents are mapped to specific prosodic constituents. The way the mapping from

syntax to phonology itself works only derives relative ranks of prosodic boundaries. The present theory is then incompatible with one of the standard assumptions in the theory of prosodic phonology. The assumption at the heart of this theory is summarized by Selkirk (1986, 385) as follows:

The proposal is that the relation between syntactic structure and prosodic structure above the foot and below the intonational phrase is defined in terms of the *ends* of syntactic constituents of designated types. [...] The idea is that a unit of phonological structure, a derived domain, will have as its terminal string the stretch of the surface syntactic structure that is demarcated by the right or left ends of selected constituents, as in (19):

- (19) a. $\alpha[\dots, \text{where } \dots \text{ contains no } \alpha[$
 b. $\dots]_{\alpha}, \text{where } \dots \text{ contains no }]_{\alpha}$

The general claim I am making here is that α will be drawn from the set of categories defined in X-bar theory, and that α indicates only a level (or type) in the X-bar hierarchy, which is to say that the syntax-to-prosodic structure mapping is claimed to be cross-categorical in nature.

The crucial point in Selkirk (1986) is that there are certain designated syntactic constituents that end up providing the edges of prosodic domains with a certain label. This notion of designated syntactic categories for prosodic domains was expressed in Selkirk (1996, 444): as follows:

- (19) The edge-based theory of the syntax-prosody interface
 Right/Left edge of $\alpha \rightarrow$ edge of β ,
 α is a syntactic category, β is a prosodic category.

The particular list of prosodic domains and how they are ‘syntactically grounded’, i.e. which syntactic nodes correspond to them, was updated in Selkirk (to appear).

Following Dresher (1994) and other works, the proposal in chapter 2 does *not* assume designated categories. The claim is that rather than specifying what type of

syntactic constituents are mapped to a single phonological phrase only the relative rank is derived. This section reviews some of the evidence bearing on the issue.

4.3.2 Phonological Rule Application

In the theory of prosodic phonology, prosodic domains are labeled and fall into several different categories.⁵ Some of the evidence used to motivate these prosodic categories is summarized below:

- (20) Evidence for this type of approach according to Inkelas (1990)
- a. Fewer Rule Domains than there are Rules. There is a small, fixed set of domains.
 - b. Exhaustive Parsing: Every utterance seems to be exhaustively parsed in those domains. Evidence: for each rule, every part of an utterance will be parsed into some domain where that rule applies.
 - c. Strict Layering: Each domain seems to be properly contained in a domain of the next higher level.

The mapping algorithm developed here also derives a strictly layered representation, but it fails to assign categories to the different grid lines. So far I have discussed evidence from phonetics that suggests that a relative unlabeled grid may actually provide sufficient information to account for the implementation. But evidence from phonological rules might eventually force us to distinguish prosodic categories of different types.

If it is true that certain phonological rules apply in prosodic domains that bear a particular label, then our representation has to be enriched to capture this fact. A second type of evidence for identifying particular types of prosodic constituents is by the presence/absence and type of boundary tone and presence/absence of a special nuclear accent (cf. Pierrehumbert (1980), Beckman and Pierrehumbert (1986), Pierrehumbert and Beckman (1988) and also the transcription system ToBI

⁵The precise number and label of categories varies between the papers.

Silverman et al. (1992)). There is a close correlation between domains of phrasal rule application and tonal domains (i.a. Jun, 1998), which supports the view that prosodic constituents that are special exist.

The representation used here is *compatible* with a mapping of the output of the algorithm to a fixed hierarchy of categorical prosodic categories. For example, one could claim that the mapping principles give us the unlabeled grid, which is then mapped to a labeled grid in a second stage. The prosodic hierarchy would then be the way the relative boundary distinctions provided by the syntax-phonology mapping are formally implemented by the phonological component.

What the theory presented here is *not* compatible with is that specific syntactic constituents are mapped to specific prosodic constituents. The way the mapping from syntax to phonology itself works only derives relative ranks of prosodic boundaries. The present theory is then incompatible with one of the standard assumptions in the theory of prosodic phonology. The idea is that rather than specifying what type of syntactic constituents are mapped to a single phonological phrase, only the relative rank is derived.

Let's assume that phonological criteria allow us to identify a prosodic domain that is the phonological phrase (Φ). The idea is now that in implementing the relational grid, the theory allows us to *choose* a line i in the grid that counts as the Φ -line. Within each foot at this line (and by implication all lower feet contained in it) the phonological rules associated with Φ will apply.

The segmental rules that pertain to the set of rules that apply in Φ can then be used to distinguish this domain from higher domains—but they no longer give information about the precise syntactic status of the elements they contain.

This theory makes predictions that are quite different from theories operating with designated syntactic nodes for specific prosodic domains. To illustrate this, I will use the segmental rule of flapping in English (thanks to Charles Reiss for suggesting this). Flapping can occur in coordinate structures:

(21) a car or a rat?

First, it predicts that two constituents A, B that are usually mapped to a single prosodic domain P_i should be able to phrase separately if we make sure that either A or B contains a prosodic break itself. This prosodic break is predicted by the present theory to be prosodically weaker compared to the one separating A and B. The need to implement this difference in boundary rank should enable the grammar to map A and B to separate domains:

(22) a cat[?] or (a rar and a hat[?].)

Conversely, If a rule does *not* apply between two constituents, it should be possible to still enable the rule to apply by further embedding the structure in such a way that the boundary separating the two constituents has to be relatively weaker compared to some yet higher boundary.

(23) a dog and (a car or (a rar and a hat[?].)

Testing these predictions would necessitate running a controlled experiment, but certainly for the rule of flapping in English the hypothesis is at least plausible. Flapping, glottalization, and glottalling interact with prosodic structure, both in American and British English (cf. Kahn, 1976, Gussenhoven, 1986):

- (24) a. American English Flapping and Glottalization:
 (a car or a rat[?]) and a hat[?]. *vs.* a cat[?](and a rar or a hat[?].)
 b. British English Glottaling:
 (a cat or a raʔ) and a haʔ. *vs.* a caʔ(and a rat or a haʔ)

In the American English example flapping occurs between lower boundaries but not higher boundaries. In the British English example glottalization occurs before higher but not lower boundaries. Prosodically conditioned processes can be used to encode prosodic ranks.

But the choice of the line of the grid within which flapping occurs is not determined by grammar. There is a possible rendition of (a) in which flapping occurs in the entire utterance. It is this kind of variation in what line is picked as the relative line for

certain rules to apply which is unexpected under theories that employ designated syntactic constituents to delimit prosodic domains; it is expected under the relational view.

The theory here is still able to capture some of the generalizations that were the motivation for the prosodic hierarchy. For example, if a rule applies within a domain P_i it necessarily occurs within all domains P_j , $j < i$. However, it assumes a greater flexibility with respect to how boundaries of a certain strength relate to syntactic nodes compared to theories that assume designated syntactic categories. This greater flexibility is indeed attested for rules such as flapping in English.

The two predictions of the relational rule can be summarized as follows: (i) when in a structure two constituents phrase together and form a single domain for some phonological rule that applies in ϕ , then by making one of them more complex and thus requiring a weaker boundary to be realized within this constituent they can be forced into separate phrases. Such effects are amply discussed in the literature, often under the heading ‘branchingness’, e.g. Zec and Inkelas (1990), Dresher (1994), Elordieta et al. (2003), and Prieto (forthcoming). The second prediction is that (ii), when two constituents do not phrase together in a single phrase ϕ , then by increasing the speech rate and/or embedding it deeper into a structure such that the boundary between the two constituents must be realized as being weaker than some other boundary, the two phrases should be able to phrase together.

The question is then how well these predictions are confirmed by empirical data. In order to test this in detail in the classic cases discussed in the literature on prosodic phonology, e.g. data from Chi Mwi:ni (Kisseberth and Abasheikh, 1974, Selkirk, 1986), Ewe (Clements, 1978, Selkirk, 1986), or Xiamen Chinese (Chen, 1987, Selkirk, 1986), more evidence with different numbers of levels of embedding and phrasings at different speech rates would be needed than are available in the original papers. I will not discuss the existing data here for reasons of time and space. The issue has to remain unresolved at this point, and future research has to bring to light on how closely syntactic and prosodic categories are tied together.

It is of interest in this context that some of the segmental rules that were originally taken to be categorical and used to define prosodic domains have turned out to be gradient once measurements were taken instead of using impressionistic evidence. This suggests that they cannot be used as evidence for designated categories since, on the contrary, they quantitatively distinguish degrees of boundary strengths. E.g., Esposito and Truckenbrodt (1998) reports that there at least two degrees of lengthening involved in *raddoppiamento sintattico*, a phonological rule in Florentine Italian that has played a big role in the shaping of theories on prosody—but crucially based on the assumption that it is a categorical test for a single type of prosodic domain, the phonological phrase. Similarly, gradient results were found in a yet unpublished study on *raddoppiamento sintattico* conducted by Mirco Ghini (p.c.).

In theories that operate with designated syntactic categories, (Selkirk, 1986, Chen, 1987, Hale and Selkirk, 1987, Selkirk, 1995b, Truckenbrodt, 1995, 1999), each conjunct in a coordinate structure involving names should be mapped to the prosodic category corresponding to XPs (maybe by XP-alignment). Say the designated category is the phonological phrase, Φ . But this theory runs into problems in capturing the type of data pattern observed in chapter 2. The predicted phrasing for *every* coordinate structure would be flat:

(25) * (Lysander) $_{\Phi}$ (and Hermia) $_{\Phi}$ (or Demetrius.) $_{\Phi}$

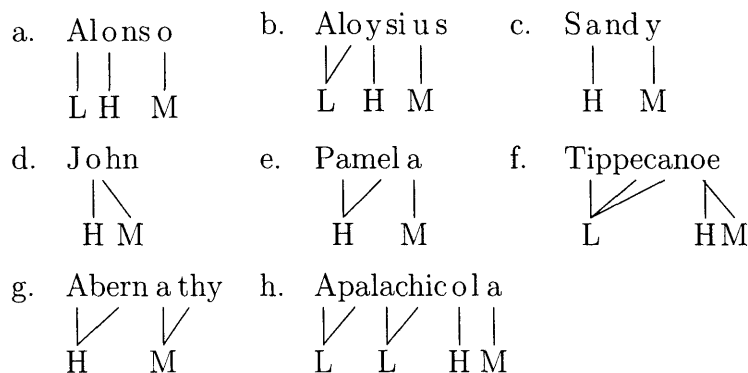
The hierarchical prosody that is actually observed constitutes ‘XP-within-XP’ effects. Similarly, the sentence coordinations investigated by Ladd (1988) (example (8)) and in Fery and Truckenbrodt (2004) constitute ‘Sentence-within-sentence’ effects. A hierarchical prosody is observed, although the elements are each of the *same* syntactic category, and hence should map to the same prosodic category. These Russian Doll effects point to a more recursive view of prosody, as it is pursued in Ladd (1986), Kubozono (1989, 1992), Ladd (1988, 1996) and Dresher (1994).

4.3.3 Tone Domains and Boundary Tones

How do the derived metrical grid structures relate to tonal elements such as boundary tones and pitch accents? I assume with Liberman (1975) and Gussenhoven (1990), Gussenhoven and Rietveld (1992) that tones are mapped to the metrical grid by principles that align independent intonational elements or ‘tunes’ to the metrical structures.

Liberman (1975) uses complex tunes and their alignment to motivate this view of how accents relate to prosody. One of his examples is the vocative chant Liberman (1975, 31). The melody that it consists of involves the following sequence of tones: [L H M]. Liberman’s point is that the tune can be identified independently of the text, and should be represented separately. The alignment with the segmental material of the constituent that is chanted depends on its prosodic structure:

(26) Tune-to-Text Alignment



The generalizations about the placement of the tune are the following, according to Liberman:

- (27)
- High tone associates with main stress of the text, and with syllables intervening between main stress and the point where mid tone associates.
 - Low tone on syllables preceding the main stress—if there are any.
 - If there is a secondary stress following the main stress, mid tone associates with it, and with all following syllables.
 - If syllables following the main stress are all unstressed, the mid tone is associated with last one.

- e. If nothing follows the main stress, then that syllable is "broken" into two distinct parts, the second of which receives the mid tone.

The grammar of tune-to-text alignment will not be addressed in this chapter. But an important point can be made by considering the nature of the conventions summarized above: they do not make reference to the fact that the examples that are used are words with a particular foot structure. Liberman (1975) could just as well have used bigger syntactic objects to make his point. The same alignment principles apply (I will ignore the finer alignment of the tune with the unstressed syllables below, just as the relative prominence within the pre-nuclear domain that aligns with the L-tones):

- (28) a. Lysander and Demetrius!
 | | |
 L H M
- b. Lysander, Hermia and Demetrius!
 | | | |
 L L H M
- c. Lysander and Demetrius arrived!
 | | |
 L H M

The line that is relevant for the alignment with the intonational tune in the representation that I use here is the top line of the grid. How many lines the top line dominates depends on how complex a syntactic structure is. The theory of text-to-tune alignment was further developed in Pierrehumbert (1980).⁶

Gussenhoven (1990) and Gussenhoven and Rietveld (1992) similarly develop and test a theory of how tonal domains are constructed based on a prosodic representation. These studies assume prosodic constituents that are distinguished by labels, and yet they do not tie those labels to specific tonal correlates. Any size of prosodic constituent (foot, word, phrase) can end up providing the prominences and boundaries with which pitch accents and boundary tones are aligned. The parallel is that the top line in the derived prosodic representation is the relevant line to define tonal domains.

⁶A similar view of how tone placement relates to the grid is also at the heart of the account of African tone systems in Purnell (1997).

To summarize, there is some evidence that intonational tunes are not tied to particular lines in a prosodic hierarchy. Tonal elements are assigned to certain segments in the structure depending on where they link in the prosodic structure that they form part of.

Embedded intonational domains with different tonal properties, as they are reported in Beckman and Pierrehumbert (1986, et seq.) and assumed in the ToBI-annotation system (Silverman et al., 1992, Beckman et al., 2005) are in principle compatible with the theory presented here. But the same reasoning that was applied to domains of rule application applies here: the prediction is that tonal domains of a particular type cannot be tied to designated syntactic categories.

Some boundary tones actually carry meaning and have been treated as ‘morphemes’ (cf. Gussenhoven (1984), Pierrehumbert and Hirschberg (1990), and Selkirk (to appear) and references therein.). They have been tied to speech act operators in Potts (2005). But then they do not give evidence for any particular prosodic unit—they simply attach to the prosodic domain they are affixed to, whichever line in the grid that may be. Similarly, sentence-final particles that encode similar meanings in many languages are not used as criteria for certain prosodic categories in those languages (although of course they might correlate with prosodic features).

That intonational patterns may really be part of the syntactic structure itself was first proposed in Stockwell (1960), who introduced phrase structure rules of the following kind: $S \rightarrow Nuc\ IP$, where *IP* stands for ‘intonational pattern’ and *Nuc* is then mapped NP VP, i.e. the actual sentence minus its intonation contour.

An updated version of this claim is to say that part of the intonation contour is the spell-out of syntactic heads of speech act operators that take sentences as their complements. Such operators were proposed and motivated in Jacobs (1991c)).

Summary

This section reviewed quantitative and qualitative evidence for prosodic structure. Quantitative durational effects at boundaries and pitch scaling effects can both be

captured by the relational grid proposed in chapter 2. Both *boundary strength scaling* and *pitch accent scaling* give evidence for relative nested grouping of prosodic domains. The observed effects differentiate prosodic domains relative to each other, but do not distinguish different categories of prosodic domains.

Qualitative distinctions of different types of domains, as are usually assumed in the literature on the ‘prosodic hierarchy’, are compatible with the presented theory. However, the relational theory is incompatible with the idea that particular syntactic categories map to particular prosodic categories. Instead, there should be flexibility in how the prosodic means are employed to realize the relative boundary ranks assigned by the syntax-to-phonology mapping. The assumption that certain syntactic constituents map to certain prosodic categories is common in the literature on prosodic phonology, but the state of the empirical evidence is not conclusive, and future inquiry will have to further investigate the issue.

One theory that seems particularly attractive in light of the evidence for recursion in coordinate structures is the one proposed in Dresher (1994, 39) based on evidence from Tiberian Hebrew. Dresher proposes that there are prosodic domains with different category labels, and motivates them based on the observation that they provide the domain for certain rhythmic and segmental phonological processes. At the same time, he presents evidence for recursion *within* those categories, and postulates that phonological rules do not have access to these within-category hierarchical distinctions. Similar within-category recursion was argued for in Ladd (1988).

So far, I have not discussed factors other than syntax that play a role in prosodic structuring. In particular, I have not discussed the role of eurhythmic principles on phrasing. For example, prosodic branchingness and principles that call for phrases of equal prosodic weight also seems to affect the choice in prosodic phrasing, as observed in Ghini (1993a,b). Eurhythmic principles are invoked in many theories of prosodic phrasing (i.a. Dresher, 1994, Selkirk, 1995b, Sandalo and Truckenbrodt, 2002, Elordieta et al., 2003, Prieto, forthcoming, Pak, 2005). Some effects of rhythm will be discussed in chapter 6, but a detailed discussion of eurhythmic effects goes beyond the scope of the present study.

